

MADHURA CHITALE

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EDUCATION

Masters in Computer Science - GPA: 4/4

Aug 2025 - May 2027

State University of New York - Stony Brook

NY, USA

Coursework: Human Computer Interaction, Data Science Fundamentals, Logic, Systems Security

B.Tech. in Computer Engineering - GPA: 3.73/4

Jul 2020 - May 2024

University of Mumbai.

Mumbai, India

Coursework: Data Structures, Operating Systems, Algorithms, System Design, Cloud Computing, Databases

TECHNICAL SKILLS

Programming Languages

: Java, Python, C, C++, JavaScript, TypeScript

Tools & Platforms

: AWS, Google Cloud Platform, Jira, Postman, Grafana, Maven, Gradle, GitHub, Tableau, Excel

Web Technologies & Frameworks : AngularJS, Spring Boot, MERN Stack, HTML, CSS

Databases : PostgreSQL, MongoDB, MySQL

Machine Learning and AI

: TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV, NLTK, Pandas, NumPy

WORK EXPERIENCE

Deutsche Bank Group | Angular, Spring Boot, PostgreSQL, GCP

Jul 2024 - Jul 2025

Software Development Engineer - I

Pune, India

- **UI/UX migration** - Migrated legacy frontend to Angular 17 by optimizing asynchronous code patterns and simplifying code logic, while reducing codebase size by **42%**
- Refactored **RESTful APIs** using **Spring Boot**, achieving reduction in response time and integrated **JUnit** and **Mockito** for Unit testing, ensuring **90% code coverage**
- Contributed to PoC and early development of an in-house **real-time collaboration** app for financial documents, projected to save **€4M/year** through version-controlled editing, validation, and seamless multi-user interaction.

IIT Bombay | AWS, Python, TensorFlow, Keras, OpenCV, NumPy, Matplotlib

Jan 2024 - Jun 2024

Research Intern (Deep Learning)

Mumbai, India

- Designed and implemented a semi-supervised deep learning framework using **Convolutional Neural Networks (CNNs)** for detecting changes in satellite radar data (SAR- Synthetic Aperture Radar) with accuracy greater than **95%**
- Integrated advanced pre-processing, feature extraction, and image analysis pipeline, enabling high-precision classification and achieving **98.78%** accuracy on standard benchmark datasets, demonstrating both robustness and scalability
- Validated model effectiveness on border surveillance data spanning **50+ km**, demonstrating applicability for large-scale monitoring and enhancing geospatial risk assessments

Deutsche Bank Group | Spring Boot, Grafana, PostgreSQL, Splunk

May 2023 - Jul 2023

Software Development Engineer Intern

Pune, India

- Engineered a **Site Reliability Engineering (SRE) dashboard** using Grafana, enabling real-time monitoring and proactive incident detection, which **reduced mean time to resolution (MTTR)** by **30%** and improved uptime
- Architected and built a usage analytics dashboard using **Grafana, Splunk, PostgreSQL, Java** to monitor **KPIs for 10+ internal applications**, capturing data on user activity, feature usage, and engagement patterns
- Enabled **drill-down analytics** to assess feature-level engagement, revealing behavior patterns that drove UI simplification and prioritized high-traffic module enhancements.

PROJECTS

Deepfake Audio Detection (DL) | Python, TensorFlow, Keras, NumPy, Matplotlib

- Engineered a deepfake audio detection system using **MFCC feature extraction** and a **hybrid CNN-LSTM model**, achieving 94% classification accuracy
- Published a research paper in **IEEE Xplore**, presented at 3rd International Conference Artificial Intelligence For Internet of Things (AIIoT) ([view paper](#))

Sentiment Analysis (ML) | Python, scikit-learn, NLTK, BERT, Twitter API, Pandas, NumPy, Matplotlib

- Developed a model for **Sentiment Analysis** and **Emotion Recognition** on 20,000+ tweets using the Twitter API, applying a classification pipeline with different algorithms including **SVM, Naive Bayes** for binary sentiment and 6-class emotion categorization
- Achieved an accuracy of **92.09% for SVM** and **90.27% for Naive Bayes** in classifying positive and negative sentiments and emotions such as happy, sad, disgust, surprise, angry and non-relevant